

Sample CS 4700/6700 1st Midterm Exam

Open books, handouts, notes, and a computer

1. (20 points)

Office Problem:

- (1) There are managers, engineers, and secretaries in an office. Each of them has an SSN and a name. In addition, each secretary has one or more skills; each engineer has an expertise, and each manager has a rank.
- (2) Each engineer and secretary is supervised by a manager, whereas each manager supervises one or more engineers and secretaries. Some manager may supervise other managers, and some manager may be supervised by other managers.
- (3) There are different software packages shared by managers, engineers and secretaries. Each software package has a unique name. Each person (i.e., manager, engineer or secretary) can use any of the software packages, and we want to keep track of who is using each software package.
- (4) There are projects performed in the office, and each project has a unique project name and a project period (composed of a start date and an end date).
- (5) Each project can involve some of the managers, engineers and secretaries. Each person is involved in one or more projects. We want to keep track of who is involved in each project and his/her weekly working hours on the project.

Draw an ER diagram describing the above problem. Specify the constraints (i.e., Cardinality ration constraint and Participation constraint) of each relationship type clearly in your ER diagram, and clearly state any assumption you make.

2. (20 points)

Convert the following ER diagram into an equivalent relational database schema diagram; that is, draw the relation schemas. Illustrate the referential integrity constraints by arrows from each foreign key to the corresponding primary key, and clearly state any assumption you make.

